# Phishing Detection & Awareness Report

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# Phishing Email Analysis Report

## 1. Report Overview

This report analyzes a suspicious email to determine whether it represents a legitimate communication or a phishing attempt. The goal was to identify indicators that could reveal malicious intent and evaluate the risk posed by the message.During the analysis, the sender information, the content of the email, the embedded links and the authentication results from the email headers are examined.

This report explains the indicators identified, how the phishing attack works and the steps users can take to avoid falling victim to similar attacks in the future

## 2. Phishing Email Overview

To begin the investigation, four phishing emails were selected for analysis. They impersonate major brands and attempt credential theft, financial fraud, or malware delivery. Raw headers are included and the email content is summarized below. I reviewed the general characteristics of the email including the sender information, subject line, and any links contained in the message.

### Tools Used

- **Google Admin Toolbox – Messageheader**
  Analyzed email headers to check SPF, DKIM, and DMARC results and identify suspicious sending servers.
- **Phishing Email Datasets**
  - SeanWrightSec Phishing Examples – Source of real phishing emails for header and content analysis.
- **Text Editor / Word Processor / PDF Software**
  Used to document findings and create the final report.

# 3. Identification of Phishing Indicators

While examining the email, several indicators that suggest the message is malicious, including:

**1. Suspicious Sender Address** -The domain does not match the legitimate domain used by the organization it claims to represent

**2. Urgent Language** - The email contains urgent language instructing the recipient to verify their account immediately.

**3. Suspicious Links** -The embedded link redirects to a website that does not belong to the official organization.

**4. Generic Greeting** -The message uses **"Dear Customer"** instead of the recipient's real name.

**5. Request for Sensitive Information**

# 4. Email Authentication Analysis

Email authentication mechanisms help verify whether a message was actually sent from the domain it claims to originate from. The three main mechanisms used for this verification are:

**SPF** - Checks if the sending server is listed in the authorized servers, if not, the SPF check fails.

**DKIM** - Determines whether the email had been digitally signed by the legitimate domain and whether the message content had been modified, if not, DKIM fails.

**DMARC** -Combines the SPF and DKIM checks to determine whether the email aligns with the domain shown in the sender's address, if not, DMARC fails.

# 4.0 Email Header Analysis & Risk Classification

## 4.1 PayPal Account Verification

**Header Highlights:**

- From: `security@paypal.com`
- Return-Path: `security@paypal.com`
- Sending Server: `smtp.badserver.ru (185.82.217.90)`
- SPF: Fail | DKIM: Fail | DMARC: Fail

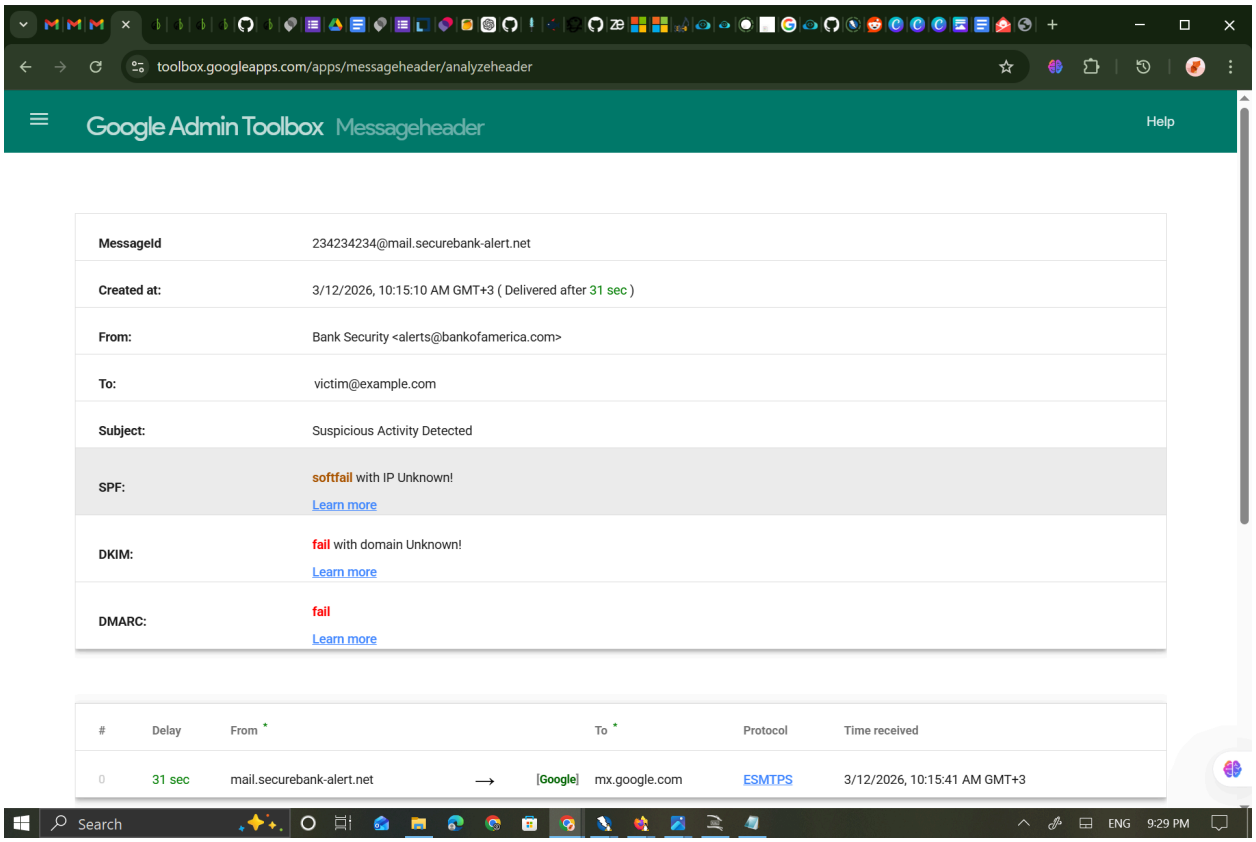
**Analysis:**
The email claims to be from PayPal, but originates from an unauthorized server. SPF and DKIM failures indicate the message is not legitimately signed by PayPal. DMARC fails because neither SPF nor DKIM align with the `From` domain.

**Risk Level:** Critical
**Reason:** Brand impersonation + multiple authentication failures + suspicious infrastructure

## 4.2 Bank Security Alert

**Header Highlights:**

- From: `alerts@bankofamerica.com`
- Sending Server: `mail.securebank-alert.net (203.190.44.12)`
- SPF: Softfail | DKIM: Fail | DMARC: Fail




**Analysis:**

The sender attempts to impersonate a bank. SPF softfail shows the IP is not fully authorized, DKIM fails, and DMARC fails due to alignment issues. The mail server is unrelated to the legitimate bank domain.

**Risk Level:** High
**Reason:** Domain impersonation + authentication failures + unrecognized mail server

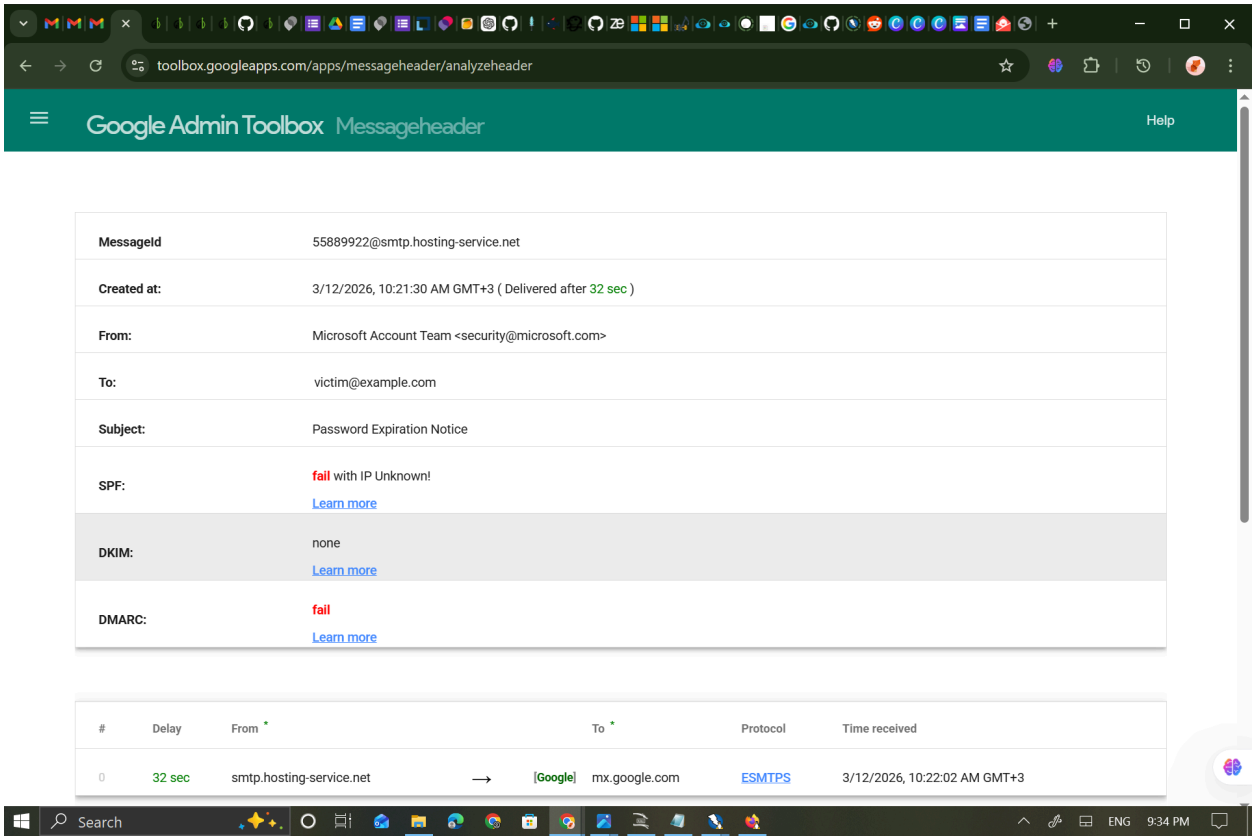
## 4.3 Microsoft Password Reset

**Header Highlights:**

- From: `security@microsoft.com`
- Sending Server: `smtp.hosting-service.net (92.118.160.15)`
- SPF: Fail | DKIM: None | DMARC: Fail

**Analysis:**
Although the email appears to come from Microsoft, it originates from a third-party hosting server. SPF fails because the IP is unauthorized, DKIM is missing, and DMARC fails.

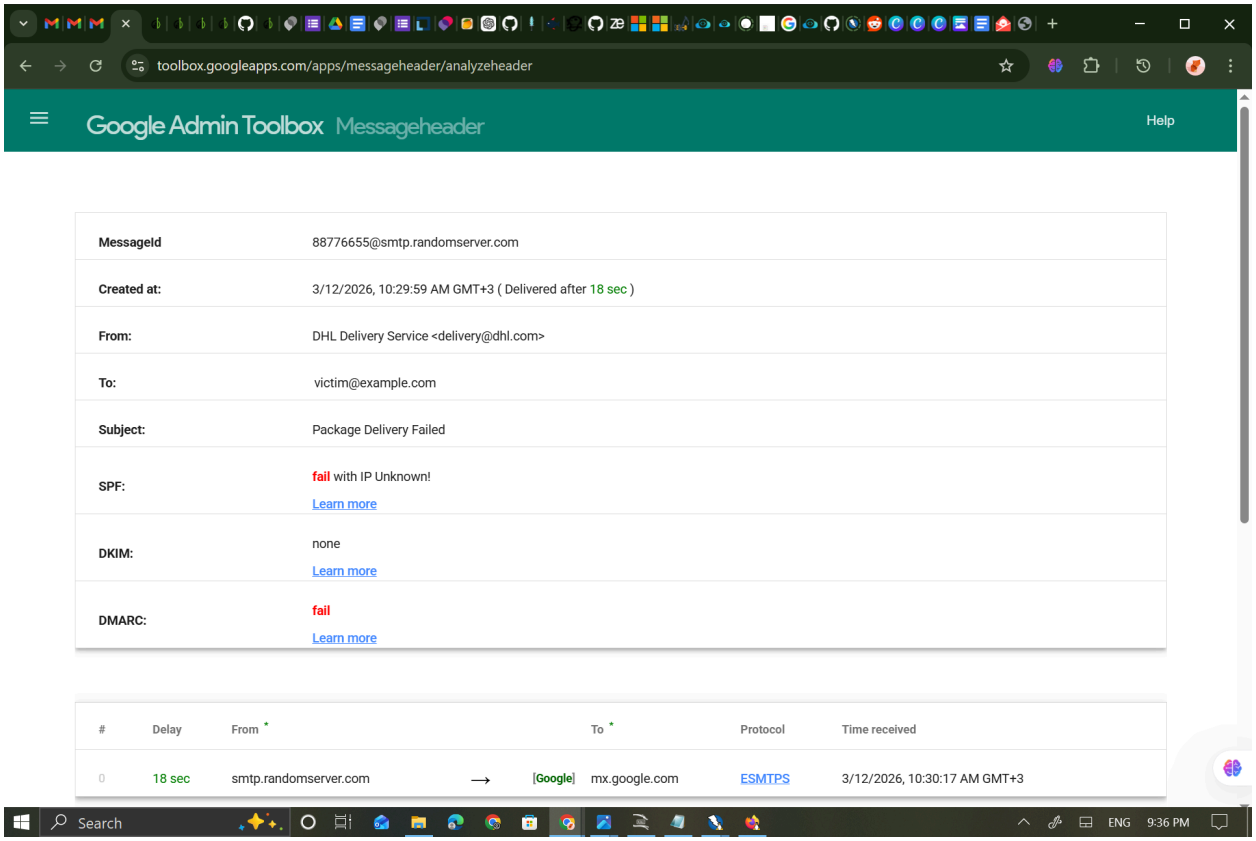
**Risk Level:** High
**Reason:** Unauthorized server + missing DKIM + failed authentication alignment




# 4.4 DHL Delivery Notification

**Header Highlights:**

- From: `delivery@dhl.com`
- Sending Server: `smtp.randomserver.com (176.44.90.55)`
- SPF: Fail | DKIM: None | DMARC: Fail




**Analysis:**

This email impersonates DHL. The sending server does not belong to DHL infrastructure. SPF and DMARC fail, and DKIM is absent, confirming the email is suspicious.

**Risk Level:** High
**Reason:** Brand impersonation + authentication failures + suspicious mail server

# 5. How the Attack Works

Based on the analysis, the attack follows a typical phishing strategy designed to steal user credentials.

First, the attacker sends an email that appears to come from a trusted organization. The message then pressures the recipient to take immediate action, such as verifying their account.

If the recipient clicks the link, they are redirected to a fraudulent website designed to look like the legitimate login page. When the victim enters their credentials, the attacker captures the information and can use it to gain unauthorized access to the victim's account.

This method is known as **credential harvesting**, a common technique used in phishing campaigns.

# 6. Prevention Recommendations

Based on the findings from this analysis, users should follow these security practices to avoid phishing attacks:

- Always verify the sender's email address before responding to a message.
- Hover over links to confirm the actual destination before clicking.
- Avoid downloading attachments from unknown sources.
- Access important services directly through official websites instead of email links.
- Report suspicious emails to the IT or cybersecurity team immediately.

Organizations should also implement email filtering systems and encourage the use of multi-factor authentication to reduce the impact of compromised credential